

Daily egg consumption in hyperlipidemic adults - Effects on endothelial function and cardiovascular risk

Background Limiting consumption of eggs, which are high in cholesterol, is generally recommended to reduce risk of cardiovascular disease. However, recent evidence suggests that dietary cholesterol has limited influence on serum cholesterol or cardiac risk. **Objective** To assess the effects of egg consumption on endothelial function and serum lipids in hyperlipidemic adults. **Methods** Randomized, placebo-controlled crossover trial of 40 hyperlipidemic adults (24 women, 16 men; average age = 59.9 ± 9.6 years; weight = 76.3 ± 21.8 kilograms; total cholesterol = 244 ± 24 mg/dL). In the acute phase, participants were randomly assigned to one of the two sequences of a single dose of three medium hardboiled eggs and a sausage/cheese breakfast sandwich. In the sustained phase, participants were then randomly assigned to one of the two sequences of two medium hardboiled eggs and 1/2 cup of egg substitute daily for six weeks. Each treatment assignment was separated by a four-week washout period. Outcome measures of interest were endothelial function measured as flow mediated dilatation (FMD) and lipid panel. **Results** Single dose egg consumption had no effects on endothelial function as compared to sausage/cheese (0.4 ± 1.9 vs. $0.4 \pm 2.4\%$; $p = 0.99$). Daily consumption of egg substitute for 6 weeks significantly improved endothelial function as compared to egg ($1.0 \pm 1.2\%$ vs. $-0.1 \pm 1.5\%$; $p < 0.01$) and lowered serum total cholesterol (-18 ± 18 vs. -5 ± 21 mg/dL; $p < 0.01$) and LDL (-14 ± 20 vs. -2 ± 19 mg/dL; $p = 0.01$). Study results (positive or negative) are expressed in terms of change relative to baseline. **Conclusions** Egg consumption was found to be non-detrimental to endothelial function and serum lipids in hyperlipidemic adults, while egg substitute consumption was beneficial.